

Thick Origami

Lab Report

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Abstract

This project investigated the design and fabrication of a thick-panel origami structure using the Bennett linkage model. Unlike zero-thickness origami, thick rigid panels cannot fold about hinge lines that meet at a single point without interference, so hinge axes must be offset to preserve compatible motion. Using the assigned sector angles $\alpha_{12} = 30^\circ$, $\alpha_{23} = 90^\circ$, $\alpha_{34} = 150^\circ$, and $\alpha_{41} = 90^\circ$, together with the prescribed offset value $a_{12} = 5 \text{ mm}$, the remaining Bennett-linkage offsets were calculated as $a_{23} = 10 \text{ mm}$, $a_{34} = 5 \text{ mm}$, and $a_{41} = 10 \text{ mm}$. These values were implemented in a Siemens NX CAD model of a four-panel single-vertex thick origami specimen with a 50 mm outer radius. To prevent panel interference during folding, the final design used a non-uniform thickness distribution, with two of the four panels also having non-uniform thickness in the form of a cutout to accommodate the folding panels. The geometry was then fabricated as a rigid-panel specimen and assembled using masking tape as hinges. The final Bennett-linkage model exhibited smooth single-degree-of-freedom folding motion, demonstrating that the offset-based design successfully reproduced the intended origami kinematics in a thick-panel form. As an extension, a Miura-Ori base unit was tiled using three repeated units and assembled into a larger folding specimen. Overall, the results show that Bennett linkage theory provides an effective basis for designing foldable thick-panel origami structures and that the same design logic can be extended from a single-vertex mechanism to a repeated deployable pattern.

1. Introduction

Origami-inspired deployable structures are attractive for engineering applications because they can achieve large shape changes through geometric motion while remaining lightweight and compact in the stowed state. In conventional zero-thickness origami, fold lines meet at a single vertex and the panels can rotate without interference. For thick rigid panels, however, the same geometry cannot be used directly because neighboring panels would collide during folding. To overcome this issue, the Bennett linkage kinematic model is used to offset the hinge axes by prescribed distances while preserving the folding motion of the original zero-thickness vertex. In this way, thick-panel origami can reproduce the kinematics of an ideal origami mechanism while accommodating real panel thickness. This concept is frequently applied to deployable aerospace structures, such as compactly stowed solar arrays that must deploy reliably in orbit.

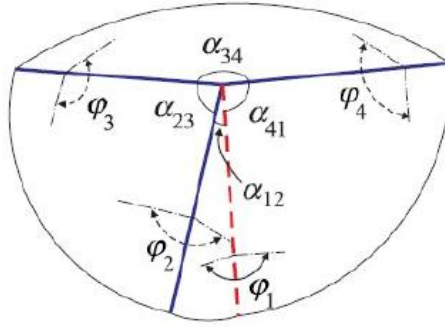


Figure 1. Zero-thickness origami vertex.

The primary objective of this project was to design and fabricate a thick-panel single-vertex origami specimen using the Bennett linkage equations as the basis for hinge-offset placement. The assigned sector angles were $\alpha_{12} = 30^\circ$, $\alpha_{23} = 90^\circ$, $\alpha_{34} = 150^\circ$, and $\alpha_{41} = 90^\circ$, shown in Figure 1, with a prescribed offset value of $a_{12} = 5 \text{ mm}$, Figure 2. Using these given parameters, the remaining offsets were determined from the Bennett linkage relations and then implemented in CAD to create a physically foldable thick-origami model. The resulting design was verified in assembly, fabricated as rigid panels by 3D printing, and connected using tape hinges to evaluate whether the structure exhibited the expected single-degree-of-freedom (1-DOF) folding behavior.

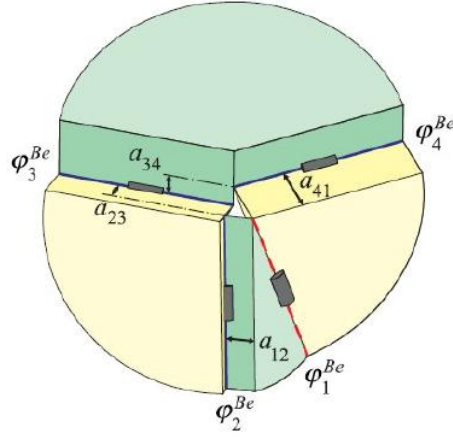


Figure 2. Thick-panel Bennett linkage where fold lines are offset by distances a_{ij} .

In addition to the main Bennett-linkage specimen, a thick Miura-Ori configuration is also designed, fabricated, and studied. This added specimen was a separate project that does not utilize the Bennett-linkage equations extensively, but served as a useful demonstration of how a compatible thick single-vertex unit can be repeated and assembled into a larger deployable pattern. In this part of the work, the single-vertex thick-origami concept was tiled in CAD to form a Miura-Ori based arrangement, and a final printed specimen was assembled to examine its folding behavior at a larger patterned scale. Because this Miura-Ori model was an extension of the main design, it will be discussed more briefly than the Bennett-linkage specimen in the sections that follow.

Overall, the purpose of this experiment was to establish a kinematic design, model the design in CAD, fabricate the specimen, and analyze the physical folding behavior in a thick-origami system. By starting from Bennett linkage equations and carrying the design through printing and taped assembly, the project evaluates whether the mathematical hinge-offset conditions produce a physically realizable rigid-panel mechanism. The additional Miura-Ori extension further illustrates how the same design logic can be scaled from a single-vertex prototype to a larger repeating origami pattern.

2. Methods

The primary specimen in this project was a thick-panel single-vertex origami structure designed using the Bennett linkage model. In contrast to zero-thickness origami, where all fold lines intersect at a single point, a thick-panel mechanism requires the hinge axes to be offset so that neighboring rigid panels do not interfere during folding. The Bennett linkage equations were therefore used as the geometric basis for the panel layout and hinge placement. According to the lab handout, the relevant conditions are the flat-foldability condition,

$$\alpha_{12} + \alpha_{34} = 180^\circ, \quad \alpha_{23} + \alpha_{41} = 180^\circ$$

the distance symmetry condition,

$$a_{12} = a_{34}, \quad a_{23} = a_{41}$$

and the Bennett ratio,

$$\frac{a_{12}}{a_{23}} = \frac{\sin(\alpha_{12})}{\sin(\alpha_{23})}$$

where α_{ij} is the sector angle between adjacent panels and a_{ij} is the offset distance between adjacent hinge axes. For this project, the assigned sector angles were $\alpha_{12} = 30^\circ$, $\alpha_{23} = 90^\circ$, $\alpha_{34} = 150^\circ$, and $\alpha_{41} = 90^\circ$, with a prescribed offset value of $a_{12} = 5 \text{ mm}$. These values satisfy flat-foldability as shown below.

$$\alpha_{12} + \alpha_{34} = 30^\circ + 150^\circ = 180^\circ$$

$$\alpha_{23} + \alpha_{41} = 90^\circ + 90^\circ = 180^\circ$$

Using the Bennett ratio, we can obtain a_{23} when $a_{12} = 5 \text{ mm}$.

$$\frac{5}{a_{23}} = \frac{\sin(30^\circ)}{\sin(90^\circ)} = \frac{0.5}{1}$$

$$a_{23} = 10 \text{ mm}$$

Thus, the final hinge-offset values used in the design were shown below.

$$a_{12} = 5 \text{ mm}, \quad a_{23} = 10 \text{ mm}, \quad a_{34} = 5 \text{ mm}, \quad a_{41} = 10 \text{ mm}$$

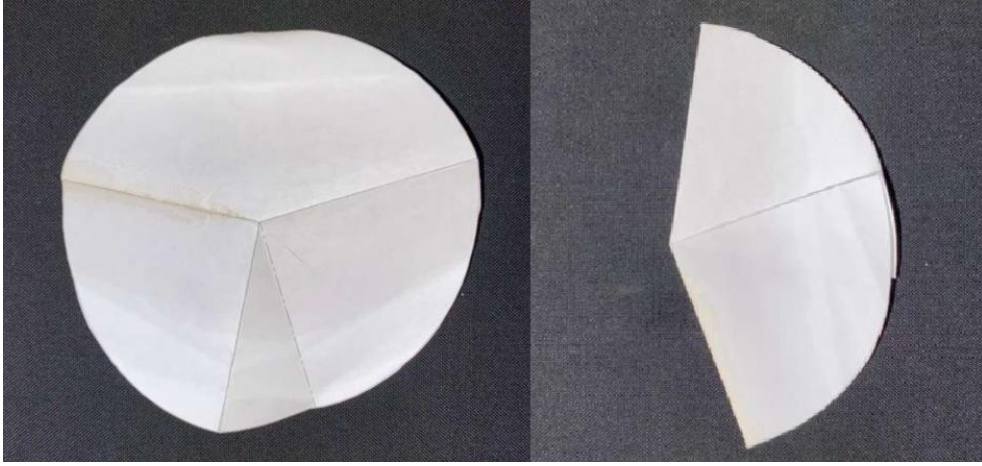


Figure 3. Zero-thickness paper pattern. Left: deployed. Right: Folded.

Before designing the thick-panel model, a pre-cut paper pattern of the corresponding four-panel single-vertex origami was folded to observe its kinematic behavior, Figure 3. This step was used to examine how the panels moved relative to one another and to verify that the underlying origami mechanism behaved as a single degree of freedom (1 DOF) system. This paper-pattern exploration is done to build intuition for the coupled panel motion before implementing the hinge offsets required for the thick origami.

Using the calculated offset values, four wedge-shaped rigid panels were modeled in Siemens NX to form the thick-panel single-vertex origami CAD design, Figure 4. Each panel was modeled with its assigned sector angle, and an appropriate outer radius was determined, using the 50 mm recommended value. The hinge-axis locations were then positioned so that the perpendicular distances between adjacent hinge axes matched the calculated values a_{12} , a_{23} , a_{34} , and a_{41} . These perpendicular distances are important because at this stage, the thick-panel geometry was no longer defined only by the wedge angles, but also by the offset placement.

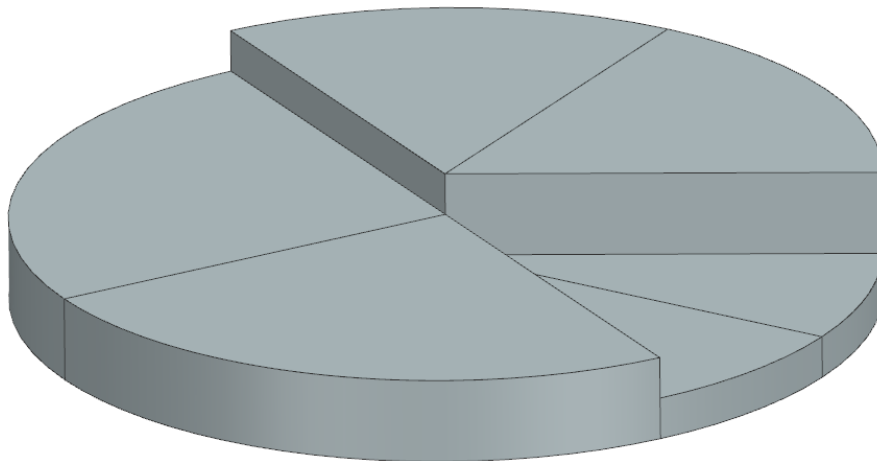


Figure 4. CAD drawing of the assembled Bennett-linkage specimen.

The panel geometry was not created with a single uniform thickness for all four panels. Instead, the thickness distribution was chosen to remain compatible with the Bennett-linkage offset configuration and to eliminate the likelihood of panel collision during folding. As a result, the four panels did not all have the same thickness, and two of the four panels were also non-uniform in thickness, as shown in Figure 5. This geometry followed the provided design diagram and the Bennett-linkage results so that the physical thick-panel layout could reproduce the intended folding kinematics while avoiding interference that would occur if all panels were given identical thickness. These modifications were necessary because a zero-thickness-inspired geometry would otherwise cause overlap or collision once rigid panel thickness was introduced.

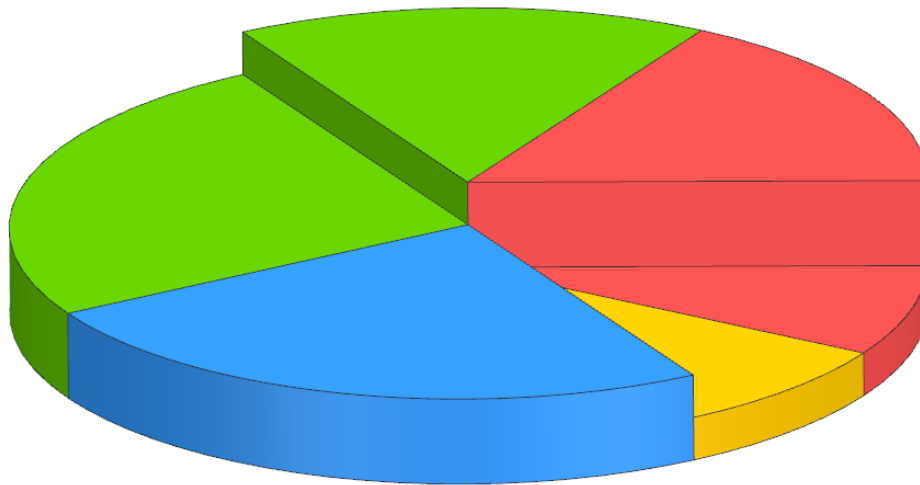


Figure 5. Color-coded CAD drawing of the assembled Bennett-linkage specimen.

Before fabrication, the four panels were assembled in CAD in the deployed configuration to verify geometric compatibility. Adjacent panel edges were aligned where the hinges would later be placed, and the perpendicular distances between adjacent hinge axes were checked against the calculated Bennett-linkage offsets. If the measured distances did not match the required values, the CAD geometry would need to be revised before fabrication.

After the CAD was finalized, the panel geometries were fabricated as rigid parts by 3D printing. The four CAD models were exported as STL files, sliced, and printed before physical assembly. This procedure was done by Nikhil, so the specific print settings were not recorded. These are not central to the purpose of the lab report, so they are not discussed further.

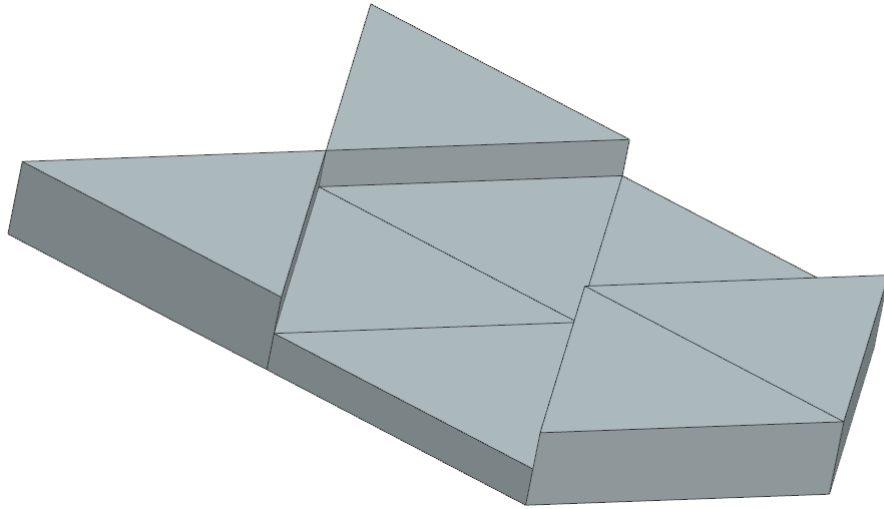


Figure 6. CAD drawing of the assembled Miura-Ori base unit.

In addition to the primary Bennett-linkage specimen, the thick-origami design was extended into a Miura-Ori patterned configuration. This extension was developed to study the mechanism of repeating the compatible thick-origami unit into a larger tiled arrangement, so that the single-vertex design could be applied to a multi-unit deployable structure. In this part of the project, the focus was not on repeating the full Bennett-linkage derivation for each unit, but rather on showing how the already established thick Miura-Ori geometry could be combined and arranged into a larger system of the same pattern. The Miura-Ori CAD base unit design was created in Siemens NX as shown in Figure 6. This step also verifies that the single thick-origami module could be extended beyond a single-vertex prototype into a larger repeating structure, Figure 7. Three base units were also 3D printed by Nikhil so that it can be assembled into a single degree of freedom system to demonstrate its folding mechanism.

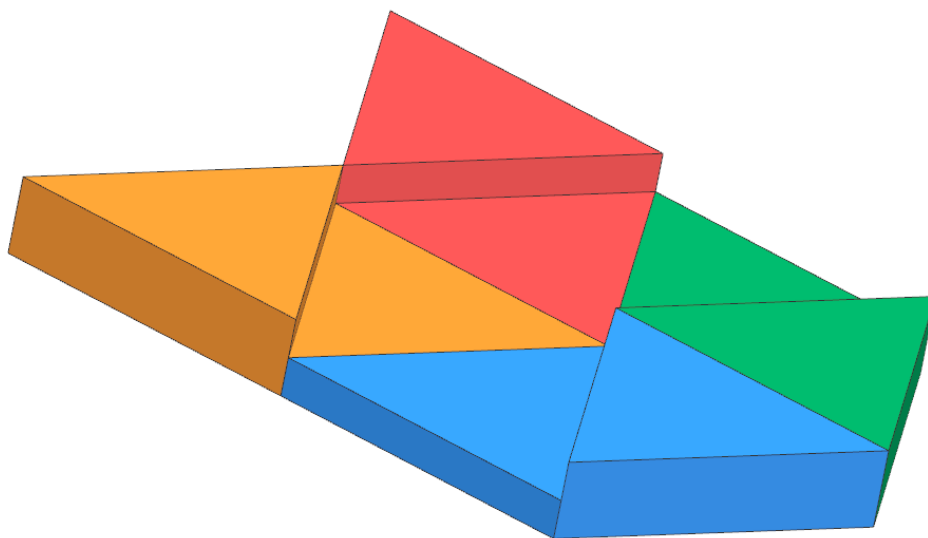


Figure 7. Color-coded CAD drawing of the assembled Miura-Ori base unit.

3. Results

The Bennett-linkage design was successfully fabricated into a physical thick-origami specimen. The printed panels reproduced the intended wedge geometry and were assembled using masking tape placed along the hinge edges to realize the prescribed hinge-axis locations, Figure 8. The placement and orientation of the tape hinges was important because it determined whether the specimen could reproduce the intended rigid-folding motion. This 3D printed assembly should demonstrate that the Bennett-linkage-based CAD design could be translated into a manufacturable specimen rather than remaining only a kinematic construction in CAD.

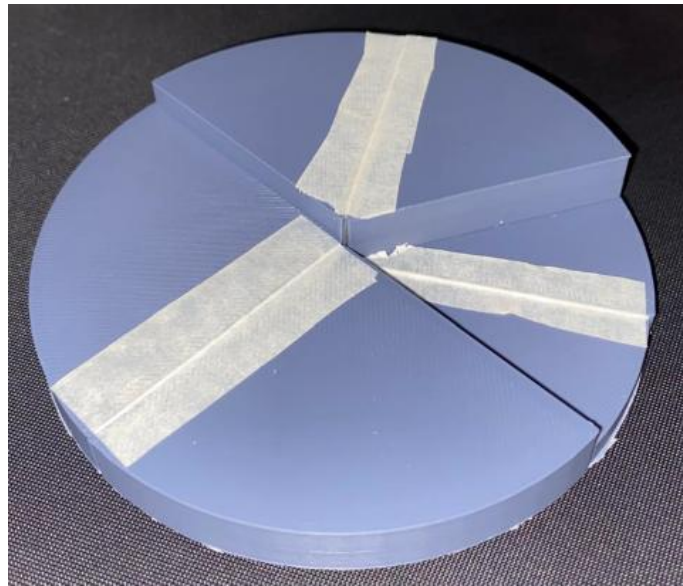


Figure 8. Fully assembled 3D-printed Bennett-linkage specimen.

The Bennett-linkage assembly successfully exhibited smooth single-degree-of-freedom folding motion. When the structure was actuated manually, motion introduced at one panel produced coordinated motion in the remaining panels, indicating that the final hinge-offset geometry and taped assembly were compatible with the intended rigid-folding kinematics. This is the central physical validation of the Bennett-linkage design, since single degree of freedom behavior is the desired outcome of this thick-origami assembly.

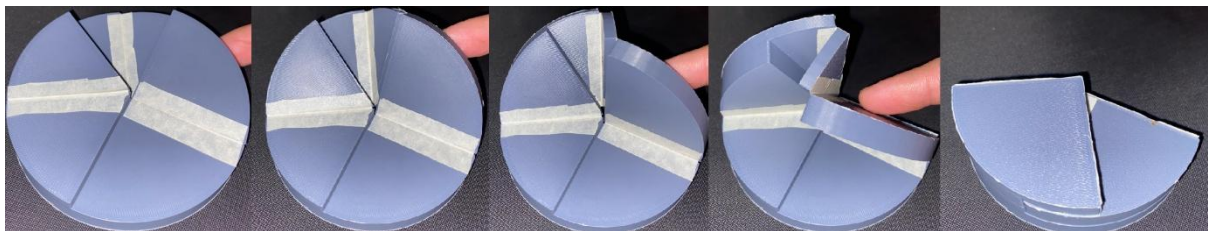


Figure 9. Folding mechanism of the 3D-printed specimen.

This result is only possible because the four panels did not have equal thickness. Instead, the smooth folding behavior was obtained only after implementing the offset-based thick-panel layout and the non-uniform thickness distribution used to prevent interference, which are modifications that had to be done to ensure that the panels does not overlap and collide with each other. In that sense, the observed motion confirms not only that the Bennett offset equations were applied correctly, but also that the resulting geometric modifications were effective in producing a physically foldable rigid-panel mechanism as shown in figure 9.

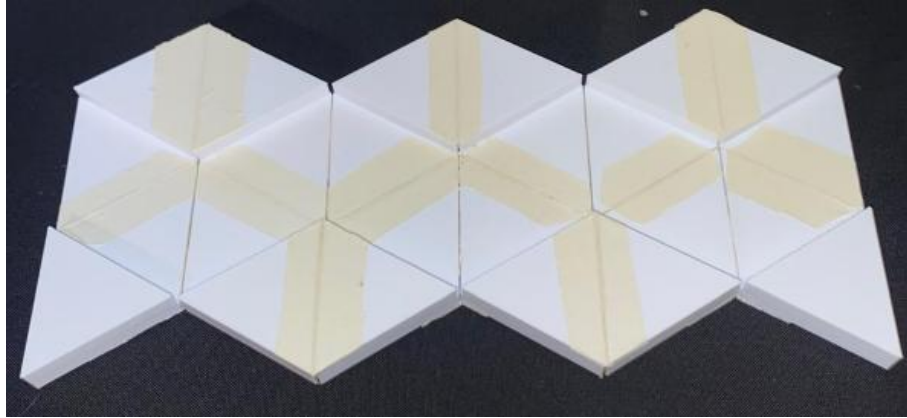


Figure 10. Fully assembled 3D-printed Miura-Ori showing three combined base units.

The Miura-Ori extension was also successfully fabricated into a physical specimen. After the base unit geometry was established in CAD, each part was printed thrice and assembled into a single connected folding structure using the same masking-tape hinge strategy as the Bennett-linkage model. The repeated units were therefore not treated as separate isolated pieces, but as one larger patterned specimen assembled from three repeated units into a continuous thick-origami configuration. This result demonstrates that the design logic developed for the single-vertex thick origami could be extended into a larger repeating layout at the specimen level.

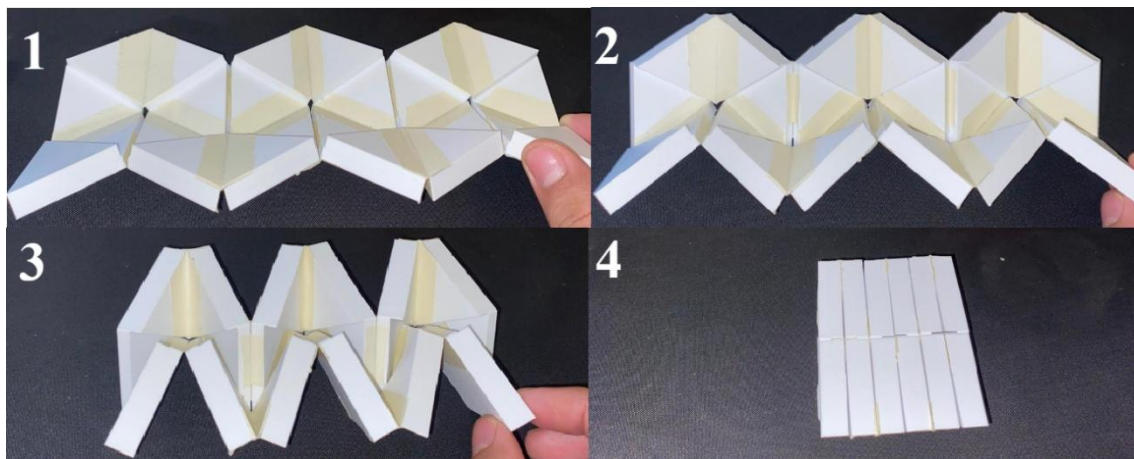


Figure 11. Folding mechanism of the 3D-printed Miura-Ori.

The Bennett-linkage specimen provided the clearest demonstration of the intended kinematic behavior, since it was observed to fold smoothly with a single dominant degree of freedom. In contrast, the Miura-Ori extension is best presented primarily as a successful design and assembly extension of the base unit into a repeated pattern, Figure 10. It shows that repeated units could be assembled into a folding thick-origami specimen using the same hinge concept as the primary model as shown in Figure 11. This indicates that the single-unit design was not limited to an isolated vertex, but could be generalized into a larger patterned structure. From an assembly standpoint, the Miura-Ori extension also served as a practical test of whether repeated units could be aligned and connected consistently using tape hinges.

The principal outcomes of the thick-origami project can therefore be summarized as follows. First, the Bennett-linkage equations produced a workable offset set that could be implemented in CAD. Second, the resulting single-vertex thick-panel model was successfully fabricated and assembled into a physical specimen. Third, the Bennett specimen demonstrated smooth single-DOF folding behavior, confirming that the calculated offsets, hinge placement, and interference-prevention geometry were physically compatible. Finally, the Miura-Ori extension showed that the same thick-origami design logic could be tiled into a larger repeated pattern and assembled into one multi-unit specimen, extending the single-vertex concept into a broader patterned configuration.

4. Conclusion

This project successfully demonstrated the design and fabrication of a thick-panel origami mechanism based on the Bennett linkage model. Using the assigned sector angles and prescribed offset value, the remaining hinge offsets were calculated and incorporated into a rigid-panel CAD design. The resulting single-vertex Bennett-linkage specimen was successfully fabricated and assembled, and the final model exhibited smooth single-degree-of-freedom folding motion. This confirmed that the Bennett linkage equations provided an effective geometric basis for translating a zero-thickness origami mechanism into a physically realizable thick-panel structure.

One of the most important findings of the project was that satisfying the Bennett-linkage equations alone was not enough to guarantee a workable physical design. In practice, interference prevention required additional geometric adaptation at the panel level. The final CAD model did not consist of four identical uniformly thick panels. Instead, the specimen used a varying thickness distribution, with two of the four panels also having non-uniform thickness in the form of cutouts to give way for the folded rigid panel. The successful folding behavior of the printed Bennett specimen therefore showed not only that the offset calculations were correct, but also that the thickness modifications were effective in preserving motion while preventing collision between panels.

The physical Bennett-linkage specimen provided the clearest validation of the design approach. It has a smooth coordinated folding which shows that the offset geometry, hinge placement, and rigid-panel layout were compatible. This verifies that the obtained offsets could be used to produce a working thick-origami folding mechanism. The final specimen therefore served as a proof of concept for rigid-panel folding based on Bennett linkage principles.

The Miura-ori extension further showed that the same thick-origami design logic could be extended beyond a single-vertex unit. By tiling the base unit in CAD and assembling three repeated units into one printed specimen, the project demonstrated that the thick single-vertex concept could be carried into a larger repeated pattern. Although this portion of the work was treated more briefly than the Bennett-linkage specimen and was not analyzed at the same level of kinematic detail, it still provided a useful extension of the main result by showing that the design could be generalized into a patterned multi-unit structure.

Overall, the thick-origami project met its main objective by showing that Bennett linkage theory can be used to design a thick rigid-panel origami structure that folds successfully in practice. The results demonstrate that a mathematically compatible hinge-offset design can be transformed into a working physical mechanism when combined with appropriate CAD-based interference prevention or modification. The additional Miura-Ori extension further suggests that this design approach can be scaled from a single-vertex prototype to a larger repeated deployable structure.